

Product Requirements Document: State Crime Intelligence Platform

Overview

This product is a secure, role-based intelligence platform for authorized law-enforcement personnel such as police officers, detectives, analysts, supervisors, and administrators. The platform combines geospatial intelligence, regional dashboards, and criminal network analysis into a single operational system designed for investigation support, situational awareness, and intelligence-led policing.[cite:3][cite:4][cite:9]

The initial release is scoped to a single state with district-level drill-down, while the architecture should support future expansion to national coverage across India. The system must prioritize controlled access, data governance, explainable analytics, and evidence-backed relationship analysis rather than acting as a generic dashboard.[cite:1][cite:4][cite:7][cite:12]

Product Vision

The platform should help law-enforcement users answer four operational questions quickly and safely:

- Who or what is active in this region right now?[cite:9][cite:11]
- How are criminals, gangs, and incidents connected?[cite:2][cite:3]
- Which areas require attention based on recent activity and intelligence signals?[cite:9][cite:12]
- What intelligence is relevant to this officer's jurisdiction and permission level?[cite:3][cite:7]

The product should reduce time spent switching across systems, improve regional and cross-district visibility, and preserve an auditable chain of intelligence access and interpretation.[cite:1][cite:3][cite:7]

Goals

Business and Mission Goals

- Create a secure digital intelligence workspace for authorized law-enforcement use.[cite:3][cite:7]
- Improve the speed of criminal intelligence discovery and regional threat review.[cite:3][cite:9]
- Support district-level and state-level operational planning through maps, dashboards, and network views.[cite:9][cite:11][cite:12]

- Enable explainable risk-zone visualization without relying on opaque AI outputs.[cite:4][cite:9]
- Provide full auditability for searches, views, and exports.[cite:1][cite:7]

User Goals

- Access only the data allowed by role, jurisdiction, and case assignment.[cite:3][cite:7]
- Understand criminal activity geographically through clear district-based visualization.[cite:11][cite:12]
- Identify relationships between individuals, gangs, vehicles, phones, addresses, and incidents.[cite:2][cite:3]
- Move from a map selection to a relevant dashboard or network investigation with context preserved.[cite:9]

Non-Goals for Version 1

- Nationwide India coverage at launch.[cite:12]
- Real-time nationwide surveillance fusion from every external agency system.[cite:1][cite:7]
- Fully autonomous predictive policing models with no human explanation layer.[cite:4][cite:7]
- Public-facing access or citizen-facing interfaces.[cite:3][cite:7]

Users and Roles

Role	Primary Use	Permissions Summary
District Officer	Regional awareness and action	View assigned districts, alerts, approved dashboards, and relevant case-linked entities.[cite:3][cite:7]
Detective	Investigation workflows	Access map, district dashboards, entity search, and network analysis for assigned or permitted investigations.[cite:3]
Intelligence Analyst	Pattern analysis	Use advanced filters, cross-district views, saved intelligence views, and risk analysis tools.[cite:2][cite:3]
Supervisor	Oversight and coordination	View broader jurisdiction summaries, workload indicators, priority trends, and team intelligence outputs.[cite:3][cite:7]
Administrator	System governance	Manage users, data-source settings, permissions, audit review, and retention controls.[cite:1][cite:7]

Core Modules

1. Access and Authorization

The platform must require secure login and allow access only to authorized users.

Authentication should support username/password plus multi-factor authentication, device or IP-based validation where appropriate, session timeout, forced reauthentication for sensitive actions, and complete login event logging.[cite:1][cite:3][cite:7]

Authorization must be role-based and, where needed, attribute-based. A user should only see the modules, records, districts, and actions permitted by role, jurisdiction, assignment, and data sensitivity.[cite:3][cite:7]

Functional Requirements

- Secure login screen with organization branding and warning banner.
- MFA for all privileged roles.
- Role-based access control.
- Record-level restrictions for sensitive subjects.
- Session timeout and reauthentication for exports, admin changes, and sensitive detail access.
- Audit logs for login, logout, failed login, permission denial, search, filter change, view access, export, and record edits.[cite:1][cite:7]

2. Geo Intelligence Map

The map is the primary situational-awareness view for the product. Version 1 must support one state with district-level boundaries and controlled zoom so the interaction remains usable and focused. The architecture must allow future expansion to a country-level hierarchy while preserving the same drill-down pattern.[cite:11][cite:12]

Map Hierarchy

- State view in Version 1.[cite:12]
- District drill-down in Version 1.[cite:11][cite:12]
- Police station, beat, or ward drill-down reserved for future versions.
- Country-level expansion reserved for later phases.

Shared Map Behaviors

- Clickable districts and state regions.
- Hover or tap summary cards with key metrics.
- Smooth zoom and pan within allowed scale limits.
- Layer switcher for map modes.
- Search for district, person, gang, vehicle, phone, address, or case.
- “Open Dashboard” and “Open Network” actions from selected context.

- Context persistence when navigating to other modules.[cite:9]

Map Mode A: Default Mode

Default mode should show the map clearly with minimal clutter, prioritizing geography and high-value operational context. It should surface district boundaries, selected-region highlight, core counters, and optional low-noise overlays such as police stations or recent incident summaries.[cite:11][cite:12]

Functional requirements:

- Clean district boundary rendering.
- Visible selected district or state highlight.
- Minimal labels at broad zoom levels.
- Optional overlays for station markers and incident counts.
- Side panel for selected region quick facts.

Map Mode B: Network Mode

Network mode should show geographically anchored relationship overlays linking criminals, gangs, incidents, and relevant entities. The view must avoid clutter by showing only meaningful links at a given zoom level and by allowing aggressive filtering.[cite:2][cite:9]

Functional requirements:

- Display lines or arcs connecting entities across districts or within a district.
- Popup or side sheet with filters for person, alias, gang, case, district, time range, relationship type, and confidence level.[cite:2][cite:3]
- Ability to isolate one criminal, one gang, or one case.
- Ability to highlight verified links only.
- “Open Detailed Network” button that redirects to the Network module while preserving current filters.[cite:9]

Map Mode C: Zone Mode

Zone mode should display a visual risk layer over the map using a transparent, explainable scoring engine rather than an unexplained black-box prediction. High-risk areas should be color-coded using a game-like intensity style, but the interface must also show what contributed to the score and how fresh the data is.[cite:4][cite:9][cite:12]

Functional requirements:

- Choropleth or grid overlay with low-to-high activity colors.
- Legend for risk scale and confidence level.
- Drill-down explanation panel showing top contributing factors.

- Time-based selector for recent, weekly, and monthly views.
- Clear distinction between confirmed incidents and modeled risk.

3. Regional Dashboard

The dashboard is opened when a user selects a state, district, or future lower-level geography and chooses the dashboard action. It should present the essential intelligence picture for that selected region using a top-to-bottom flow of KPIs, trends, priority entities, and supporting detail.[cite:9][cite:11]

Dashboard Sections

Section	Description
Region Header	Geography name, jurisdiction, current threat level, selected time range, and quick actions.[cite:9]
Key Indicators	Incidents, active gangs, wanted suspects, open cases, recent arrests, and repeat offender counts.[cite:9][cite:11]
Trend Analysis	Activity trends over 7, 30, and 90 days; incident categories; hotspot shifts.[cite:9][cite:11]
Priority Entities	High-priority criminals, gangs, repeat offenders, linked vehicles, and phones.[cite:3][cite:9]
Location Intelligence	Hot streets, wards, police station summaries, cluster movement, and recent hotspots.[cite:11][cite:12]
Intelligence Feed	Alerts, new cross-district links, resurfacing offenders, and significant changes.[cite:3]
Case and Assignment Panel	Assigned officers or units, open operations, and recent case notes if permitted.

Functional Requirements

- Dashboard loads according to selected geography from map or search.
- Filters for date range, crime type, gang, risk level, status, and confidence.
- Cards, charts, and entity lists update together when filters change.
- Ability to open a person, gang, incident, or case detail panel.
- Ability to pin items to a case workspace or export permitted summaries.

4. Network Intelligence Module

The Network module is dedicated to relationship analysis across criminals, gangs, cases, and supporting entities. The goal is not simply to display a graph but to help police discover patterns, verify relationships, and navigate supporting evidence.[cite:2][cite:3]

Page Layout

- Left panel for filters and saved views.
- Center graph canvas for the network.
- Right-side evidence drawer for selected node or edge details.[cite:2][cite:3]

Functional Requirements

- Search by person, alias, gang, phone, vehicle, address, incident, or case.[cite:2][cite:3]
- Filter by district, time range, relation type, confidence, status, and source.
- Expand and collapse neighboring nodes.
- Highlight shortest path or strongest path between two entities.
- Cluster by gang, geography, or entity type.
- Show only verified, probable, or all links based on user choice.
- Save graph view for future reference.
- Open source records and evidence explanation for each connection.[cite:2][cite:3]

Evidence and Explainability Requirements

Every node and edge should support a detail panel containing:

- Entity type and identifier.
- Aliases or alternate names where relevant.
- Associated district or region.
- Source records.
- Confidence score.
- Last updated time.
- Related incidents or cases.
- Human-readable explanation of why the connection exists.[cite:2][cite:3]

5. Case Workspace

A case workspace should be included in scope as a lightweight but important module for investigators. It provides continuity between map exploration, dashboard review, and network analysis by allowing users to pin entities, save views, and collect notes around an operational thread.[cite:3][cite:9]

Functional Requirements

- Create or select a case workspace if permission allows.
- Pin criminals, gangs, locations, incidents, and graph views.
- Save map filters and dashboard filters to the case.
- Add case notes and analyst annotations.
- Export permitted intelligence summaries with redaction where needed.[cite:7]

6. Administration and Governance

Governance controls are mandatory because criminal intelligence systems require disciplined information handling, controlled dissemination, and accountability. The administrative module must support user management, source management, retention policy controls, and audit review.[cite:1][cite:4][cite:7]

Functional Requirements

- User, role, and district assignment management.
- Sensitive tag management for high-risk records.
- Data-source registration and status monitoring.
- Retention policy controls and archival workflows.
- Audit viewer with filters by user, action, module, and time.
- Export policy and redaction rules.[cite:1][cite:7]

User Stories

Access and Security

- As a detective, access should be limited to permitted districts and assigned case data so that unauthorized records are not exposed.[cite:3][cite:7]
- As an administrator, login attempts and permission denials should be visible in audit logs so that misuse can be investigated.[cite:1][cite:7]

Map Workflows

- As a district officer, selecting a district on the map should open a concise regional intelligence overview for rapid situational awareness.[cite:9][cite:11]
- As an analyst, switching to network mode and filtering by a gang should reduce the overlay to only the relevant connections.[cite:2][cite:9]
- As a supervisor, switching to zone mode should reveal higher-risk areas with an explanation of the score drivers.[cite:4][cite:9]

Dashboard Workflows

- As a detective, the dashboard should highlight priority suspects, current incidents, and linked entities in the selected region.[cite:3][cite:9]
- As an analyst, applying a crime-type and time-range filter should update every dashboard component consistently.

Network Workflows

- As an analyst, the network module should show how a suspect is connected to other people, gangs, phones, vehicles, and incidents with evidence for each edge.[cite:2][cite:3]
- As a detective, opening a graph node should reveal source records and recent activity before any action is taken.[cite:2][cite:3]

Case Workflows

- As an investigator, it should be possible to save a filtered network view to a case workspace for later review.
- As a supervisor, permitted reports should support redaction before sharing outside the immediate team.[cite:7]

Functional Requirements by Entity Model

The system should use a common data model that supports cross-linking and graph exploration. Core entities for Version 1 are listed below.[cite:2][cite:3][cite:9]

Entity	Examples of Stored Attributes
Person/Criminal	Name, aliases, identifiers, risk level, status, known addresses, known associates, case references.
Gang/Organization	Name, aliases, hierarchy, active districts, known members, incident links.
Incident/Case	Type, date, location, severity, involved entities, status, assigned unit.
District/Zone	Name, geometry, administrative hierarchy, recent counts, risk score.
Address/Location	Coordinates, address string, linked persons, linked incidents.
Vehicle	Registration number, type, linked persons, sightings, incidents.
Phone/Device	Number or device ID, linked persons, contact patterns, related incidents.
Officer/Unit	Name, role, district, assignment, case links.
Relationship/Link Evidence	Source, relation type, confidence, last update, case reference, verification state.[cite:2][cite:3]

Relationship Types

- person to gang membership.
- person to person association.
- person to incident involvement.

- person to phone usage.
- person to address residency or visit.
- vehicle to district sightings.
- gang to district activity.
- case to entity relevance.[cite:2][cite:3]

AI and Analytics Requirements

The analytics layer should begin with explainable scoring rather than advanced black-box machine learning. The first version of Zone mode should compute region-level risk based on transparent inputs and make those inputs visible in the UI.[cite:4][cite:9]

Candidate Scoring Inputs

- Historical incident density.[cite:9][cite:11]
- Recency weighting for recent activity.[cite:9]
- Repeat-offender concentration.
- Gang activity concentration.[cite:9]
- Cross-district spillover signals.
- Time-of-day and day-of-week pattern shifts.[cite:11]

Analytics Output Requirements

- Risk score from 0 to 100.
- Confidence band or confidence label.
- Top reasons contributing to the score.
- Last refresh timestamp.
- Optional patrol priority suggestion.
- Clear disclaimer separating model output from confirmed incident counts.[cite:4][cite:9]

Data Sources and Integration Requirements

The platform should support ingesting internal and approved external intelligence sources over time, but Version 1 may start with controlled pilot datasets and mock integrations. Each record and relationship must preserve provenance so analysts can inspect source quality and timeliness.[cite:1][cite:2][cite:7]

Source Requirements

- Source identifier and system of origin on every imported record.[cite:1][cite:2]
- Ingestion timestamp and last update timestamp.
- Source confidence or trust tier.

- Deduplication and entity resolution support.
- Data quality flags for incomplete or conflicting records.[cite:1]

Security, Compliance, and Audit Requirements

Security and governance are foundational and not optional. Criminal intelligence practices emphasize controlled sharing, lawful handling, and operational accountability, so the product must expose an auditable trail across all sensitive workflows.[cite:1][cite:4][cite:7]

Security Requirements

- Encryption in transit and at rest.
- Strong authentication with MFA.[cite:3][cite:7]
- Role-based and record-level authorization.[cite:3][cite:7]
- Full audit logging for authentication, search, filtering, viewing, editing, and exporting.[cite:1][cite:7]
- Sensitive record tagging and restricted access pathways.
- Redaction support for export and inter-unit sharing.[cite:7]

Audit Requirements

Audit logs should capture:

- User identity.
- Role and jurisdiction at action time.
- Action type.
- Module used.
- Target record or filter context if applicable.
- Timestamp.
- Success or failure state.[cite:1][cite:7]

UX and UI Requirements

The product should function as a unified application shell rather than disconnected pages. A single navigation system should contain Map, Dashboard, Network, Cases, and Admin, while the selected district, gang, case, or time filter should remain available across views when appropriate.[cite:3][cite:9]

UX Principles

- Keep the default map clean and uncluttered.[cite:11][cite:12]
- Preserve user context between modules.[cite:9]
- Prioritize explanation over decoration for AI outputs and relationship lines.[cite:4][cite:7]

- Use progressive disclosure so detailed evidence appears on demand rather than overwhelming the main screen.
- Make every sensitive action explicit and reviewable.

Key UI Components

- App shell with role-aware sidebar navigation.
- Global search bar.
- Map canvas with layer controls.
- Filter drawers and saved views.
- Entity detail panel.
- Evidence panel for nodes and edges.
- KPI cards, trend charts, and ranked tables.
- Audit and admin tables.

Technical Requirements

Suggested Architecture

A practical implementation should use a modular web architecture with a strong geospatial and relationship-analysis foundation.[cite:3][cite:9][cite:12]

Layer	Recommendation
Frontend	React or Next.js web application with a role-aware app shell.
Mapping	Leaflet or Mapbox with GeoJSON district boundaries and layer controls.[cite:11][cite:12]
Network Graph	Cytoscape.js, Sigma.js, or D3-based graph visualization.
Backend API	Node.js or Spring Boot service layer.
Primary Data Store	PostgreSQL with PostGIS for geo queries.
Relationship Analysis Store	Neo4j or a graph-capable relationship service for entity connections.
Search	PostgreSQL full-text search or Elasticsearch for entity lookup.
Auth	Enterprise identity provider or custom RBAC service with MFA.
Analytics	Python service or backend jobs for scoring, deduplication, and periodic model refresh.

System Characteristics

- Scalable data model for future country expansion.
- District-level geometry handling in Version 1.[cite:12]
- Asynchronous processing for analytics refresh and relationship recomputation.
- Strong API versioning for future integrations.

- Observability for errors, ingestion health, and audit pipeline status.

APIs and Service Modules

Core API Domains

- Authentication and session service.
- User and permission service.
- Geography and map layer service.
- Dashboard aggregation service.
- Entity search service.
- Relationship graph service.
- Risk analytics service.
- Case workspace service.
- Audit and admin service.

Example API Capabilities

- Get accessible districts for a user.
- Get district dashboard summary by time range.
- Search person or gang by name or alias.
- Retrieve entity graph around a selected node.
- Retrieve risk zone layer for selected geography and time window.
- Save case workspace with pinned entities.
- Export approved intelligence summary.
- Query audit logs by user or module.

Success Metrics

Adoption and Usability

- Percentage of authorized users actively using the system weekly.
- Average time to locate a target criminal, gang, or district profile.
- Average time to move from map selection to usable regional dashboard.
- Frequency of saved views and case workspaces.

Operational Intelligence Quality

- Percentage of graph edges with traceable source evidence.[cite:2][cite:3]
- Percentage of records with source provenance and update time.[cite:1][cite:2]
- Analyst acceptance rate of risk-zone outputs after explanation review.
- Reduction in duplicate entity records after resolution workflows.

Governance and Security

- MFA adoption rate for active users.[cite:7]
- Number of unauthorized access attempts blocked.
- Audit completeness for sensitive workflows.[cite:1][cite:7]
- Time to investigate suspicious platform activity using audit logs.[cite:1]

Risks and Mitigations

Risk	Impact	Mitigation
Overly cluttered map and graph views	Users miss important signals	Use strict defaults, aggressive filtering, and progressive disclosure.[cite:9][cite:11]
Opaque AI risk scoring	Low trust and policy concerns	Use explainable scoring with visible factors and confidence labels.[cite:4][cite:7]
Unauthorized or excessive data exposure	Serious operational and legal risk	Enforce RBAC, record-level access, MFA, and audit logging. [cite:1][cite:3][cite:7]
Weak source provenance	Poor intelligence quality	Store source, freshness, and confidence for all records and links.[cite:1][cite:2]
Premature national-scale scope	Delivery delays and design complexity	Keep Version 1 limited to one state with scalable architecture.[cite:12]

Release Plan

Phase 1: Foundational Pilot

- Secure login and authorization.
- State map with district drill-down.
- Regional dashboard.
- Basic network analysis module.
- Audit logs.
- Controlled pilot dataset and explainable zone scoring baseline.[cite:3][cite:7][cite:9]

Phase 2: Intelligence Maturity

- Advanced filters and saved views.
- Case workspace enhancements.
- Cross-district alerts and intelligence feed.
- Better risk modeling and confidence controls.
- Export workflows with redaction.[cite:7][cite:9]

Phase 3: Scale and Integration

- Multi-state or national hierarchy support.
- Lower-level geographic drill-down.
- Broader source integrations.
- Deeper anomaly detection and operational recommendations.
- Inter-agency collaboration controls and deconfliction support.[cite:1][cite:7]

Open Questions

- What specific data sources will be available in the first pilot?
- Which jurisdictions or departments will participate in Version 1?
- What legal and policy constraints apply to inter-district data sharing?
- Should case workspaces allow collaborative editing from multiple units?
- Which fields require mandatory redaction in shared exports?
- What confidence framework should be used for relationship scoring?
- How often should risk-zone analytics refresh in operational use?

Recommended Next Deliverables

The best follow-up artifacts after this PRD are:

1. Database schema and entity-relationship design.
2. API specification by module.
3. Low-fidelity wireframes for Login, Map, Dashboard, Network, and Case Workspace.
4. A role-permission matrix with district-level examples.
5. A pilot rollout plan and test dataset specification.